

CHAPTER 4

Ensemble of Uncoupled Oscillators

- ❑ The optical properties of semiconductors are determined by the interaction of various types of oscillators and the electromagnetic field
- ❑ An incident electromagnetic field will cause oscillators to perform forced oscillations – depend on
 - the frequency of the incident light
 - the eigenfrequency of the oscillators
 - the coupling strength
 - the damping of the oscillation

Main content

- **4.1 Equations of Motion and the Dielectric Function Based on Classic Theory**
- **4.2 Corrections Due to Quantum Mechanics and Local Fields**
- **4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction**
- **4.4 The Spectra of Reflection and Transmission**

4.1 Equations of Motion and the Dielectric Function

- Assume that: An ensemble of identical uncoupled harmonic oscillators, neglecting damping for the moment.
- Then eigen-frequency ω'_0 :

$$\omega'_0{}^2 = \beta m^{-1} \quad (4.1)$$

- Fig.4.1b: $\lambda = \infty$ or $k = 0$ (4.2a)

- Fig.4.1c: $\lambda_{\min} = 2a$ or $k_{\max} = \frac{\pi}{a}$ (4.2b)

1. Γ -point: $k = 0$ gives the center of the first Brillouin zone
2. (4.2b): the shortest physically meaningful wavelength and the largest k -vector gives the boundary of the first Brillouin zone for a simple linear chain or cubic lattice.

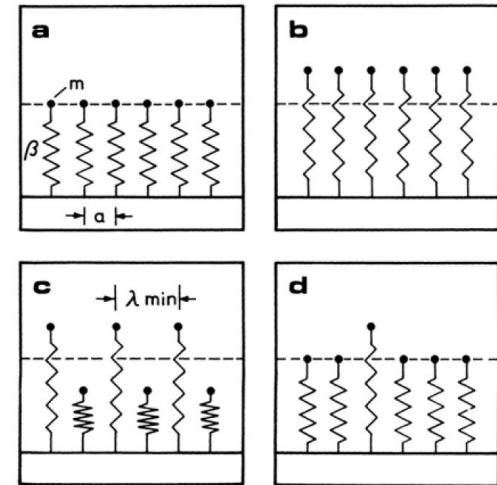


Fig. 4.1. A part of a periodic array of uncoupled oscillators in their equilibrium position (a); elongated with a wavelength $\lambda \Rightarrow \infty$ (b); with the shortest physically meaningful wavelength $\lambda_{\min}$ (c), and a non-propagating wave packet (d). After [63H1]

Uncoupled and coupled oscillators

zero coupling $\leftrightarrow$ zero bandwidth $\leftrightarrow \nu_g = 0$, Fig.4.2a

(4.3)

finite coupling $\leftrightarrow$ finite bandwidth $\leftrightarrow \nu_g \neq 0$, Fig.4.2b

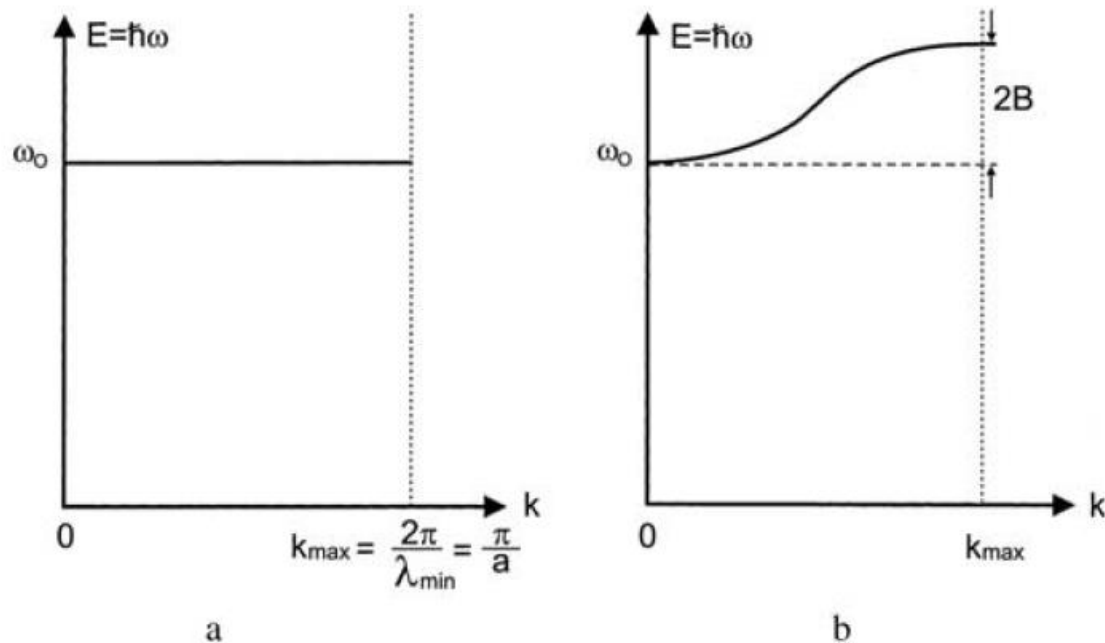


Fig. 4.2. The dispersion relation of an ensemble of uncoupled (a) and of coupled (b) oscillators

- The electric field of incident light on the independent oscillators:

$$\mathbf{E} = (E_0, 0, 0) \exp[i(k_z z - \omega t)] \quad (4.4a)$$

Considering the oscillator at $z = 0$:

$$\mathbf{E} = (E_0, 0, 0) e^{-i\omega t} \quad (4.4b)$$

Assume every oscillator carries a charge e , then a force by $\mathbf{E}$:

$$\mathbf{F} = eE_0 e^{-i\omega t} \quad (4.5)$$

The equation of motion is:

$$m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \quad (4.6)$$

The general solution of (4.6) = the general solution of the corresponding homogeneous equation + a special solution of the inhomogeneous one.

Assume : $x(t) = x_0 \exp\left[-i\left(\omega_0'^2 - \gamma^2/4\right)^{1/2} t\right] \exp(-t\gamma/2) + x_p e^{-i\omega t}$ (4.7)

For $\omega_0'^2 - \gamma^2/4 > 0$: defines the regime of weak damping.

The solution of the homogeneous equation describes a transient feature

The second part in Eq (4.7) is a forced oscillation, substitute it into (4.6):

$$x_p = \frac{eE_0}{m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} \quad (4.9)$$

The dipole moment :

$$p_x = ex_p \quad (4.10)$$

- The polarizability: $\hat{a}(\omega) = \frac{ex_p}{E_0} = \frac{e^2}{m}(\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1}$ (4.11)

- The polarization density $\mathbf{P}$ with amplitude $\mathbf{P}_0$:

$$\mathbf{P}_0 = N\hat{a}\mathbf{E}_0 = \frac{Ne^2}{m}(\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1}\mathbf{E}_0 \quad (4.12)$$

- This means that we describe the ensemble of oscillators as an effective medium, an approach which is well justified for $a \ll \lambda$
- From (4.12) and (2.27): we get for the dielectric displacement:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 \left[1 + \frac{Ne^2}{\varepsilon_0 m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} \right] \mathbf{E} \quad (4.13)$$

$$\varepsilon(\omega) = 1 + \frac{Ne^2}{\varepsilon_0 m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} = \chi(\omega) + 1 \quad (4.14)$$

4.2 Corrections Due to Quantum Mechanics and Local Fields

- In classic mechanics model, the term Ne^2m^{-1} gives the coupling strength
- In quantum mechanics, this coupling is given by the square of transition matrix element for dipole-allowed transitions :

$$|H_{ij}^D|^2 = \left| \langle j | H^D | i \rangle \right|^2 \quad (4.15)$$

- Usually define the oscillator strength : $f = \frac{2N\omega'_0}{\varepsilon_0\hbar} |H_{ij}^D|^2$ (4.16)

and

$$\varepsilon(\omega) = 1 + \frac{f}{\omega_0'^2 - \omega^2 - i\omega\gamma} \quad (4.17)$$

- For dilute systems, E is just the external incident field and use (4.17) as it is
- For dense systems, consists of two parts = external field and the field created by all the other dipoles. $\varepsilon(\omega)$ can be written :

$$\varepsilon(\omega) = 1 + \frac{f}{\omega_0^2 - \omega^2 - i\omega\gamma} \quad (4.19)$$

with a shifted eigenfrequency :

$$\omega_0^2 = \omega_0'^2 - \frac{Ne^2}{3m\varepsilon_0} = \omega_0'^2 - \frac{f}{3} \quad (4.18b)$$

4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction

- A semiconductor contains not only one type of oscillators
- Helmholtz-Ketteler formula or Kramers-Heisenberg dielectric function:

$$\varepsilon(\omega) = 1 + \sum_j \frac{f_j}{\omega_{0j}^2 - \omega^2 - i\omega\gamma_j} \quad (4.20a)$$

If the eigenfrequencies ω_0 form a continuous band:

$$\varepsilon(\omega) = 1 + \int \frac{f(\omega_0)}{\omega_0^2 - \omega^2 - i\omega\gamma(\omega_0)} d\omega_0 \quad (4.20b)$$

For simplicity, we use (4.20a) in the following.

- For a single resonance ω_{0j} well separated from all other resonances, the contribution of other resonance $\omega_{0j'}$ in the vicinity of ω_{0j}

$$\text{For } \omega \ll \omega_{0j}, \quad \frac{f_j}{\omega_{0j}^2} = \text{const} \quad (4.21)$$

and for the contribution tends to zero for $\omega \gg \omega_{0j}$,

So we can neglect the lower resonance and summarize all higher resonances in a so-called background dielectric constant ε_b

So we get the simplified expression in the surroundings of ω_{0j} :

$$\varepsilon(\omega) = \varepsilon_b + \frac{f_j}{\omega_{0j}^2 - \omega^2 - i\omega\gamma_j} \quad (4.22a)$$

- Separated into real and imaginary parts:

$$\begin{aligned}\varepsilon(\omega) &= \left(\varepsilon_b + \frac{f(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} + i \frac{\omega \gamma f}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} \right) \\ &= \varepsilon_1(\omega) + i\varepsilon_2(\omega)\end{aligned}\quad (4.22b)$$

For $\gamma \Rightarrow 0$, $(\omega_0^2 - \omega^2) = (\omega_0 + \omega)(\omega_0 - \omega) \approx 2\omega_0(\omega_0 - \omega)$, we get :

$$\varepsilon(\omega) = \varepsilon_b + \frac{f}{2\omega_0} \frac{1}{\omega_0 - \omega} \quad (4.22c)$$

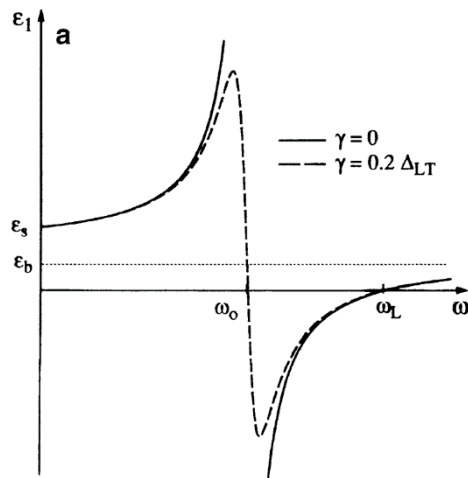
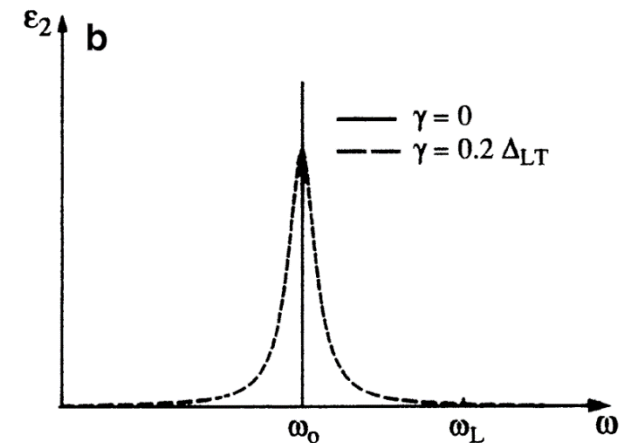


Fig. 4.3 The real (a) and imaginary (b) parts of the dielectric function for zero and finite damping



$\gamma \rightarrow 0$, there is a pole in $\text{Re}\{\varepsilon(\omega)\}$, and $\text{Im}\{\varepsilon(\omega)\}$ converges to a δ function at ω_0 . Finite damping results in a broadening of $\text{Im}\{\varepsilon(\omega)\}$ to the Lorentzian lineshape of (4.22b) and a smooth connection of the two branches of $\text{Re}\{\varepsilon(\omega)\}$.

- Small damping :

For $\gamma \Rightarrow 0$, we find: $\varepsilon(\omega = \omega_T = \omega_0) = \infty$ connected with the singularity.

$$\varepsilon(\omega = \omega_L) = 0 \quad \text{The solution of } \nabla \cdot \mathbf{D} = \varepsilon_0 \nabla \cdot \varepsilon(\omega) \cdot \mathbf{E} = 0 \quad (4.23)$$

- The relation between these two frequencies:

$$\omega_L^2 - \omega_T^2 = f / \varepsilon_b \sim |H_{ij}^D|^2 \quad (4.25)$$

$$\text{So : } |H_{ij}^D|^2 \neq 0 \Leftrightarrow f \neq 0 \Leftrightarrow \Delta_{LT} = \hbar(\omega_L - \omega_T) \neq 0 \quad (4.26)$$

See Fig.4.4 : a longitudinal electric field, acts as an additional restoring force and increases the longitudinal eigenfrequency ω_L above the transverse eigenfrequency ω_T

- The static dielectric constant ε_s below ω_0 and the value above ε_b : $\varepsilon_s = \varepsilon_b + \frac{f}{\omega_0^2}$ (4.27)
and the Lyddane-Sachs-Teller relation:

$$\frac{\varepsilon_s}{\varepsilon_b} = \frac{\omega_L^2}{\omega_0^2} > 1 \quad \text{for } f > 0 \quad (4.28)$$

- The term “small damping”:

$$\gamma < \hbar^{-1} \Delta_{LT} = \omega_L - \omega_T \quad (4.29)$$

and frequency ω far above or below ω_0 means:

$$\hbar |\omega - \omega_0| \gg \Delta_{LT} \quad (4.30)$$

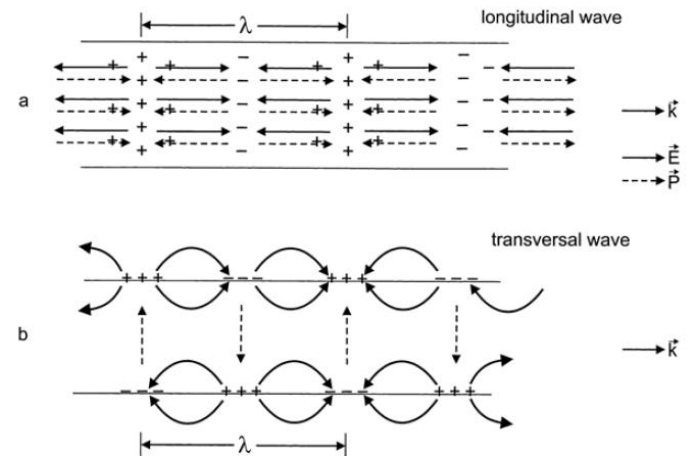
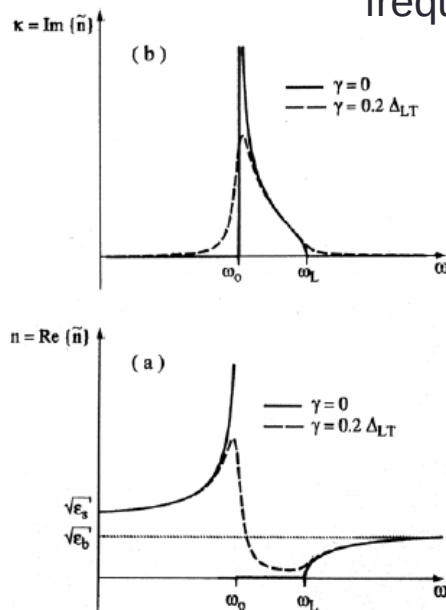


Fig. 4.4. The polarisation charges and the resulting electric and polarisation fields (solid and dashed arrows, respectively) in a slab of matter for a longitudinal (a) and a transversal wave (b)

- The complex index of refraction : $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega)$ $\tilde{n}(\omega) = \varepsilon^{1/2}(\omega)$
consequently: $\varepsilon_1(\omega) = n^2(\omega) - \kappa^2(\omega)$ $\varepsilon_2(\omega) = 2n(\omega)\kappa(\omega)$

$$\text{or : } n(\omega) = \left(\frac{1}{2} \left\{ \varepsilon_1(\omega) + [\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2} \quad \kappa(\omega) = \left(\frac{1}{2} \left\{ -\varepsilon_1(\omega) + [\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2} \quad (4.32)$$

When light frequency approaching the resonance from the low-frequency side, n increase drastically. Between ω_0 and ω_L we find



$$\omega_0 \leq \omega \leq \omega_L \quad \begin{cases} n = 0 & \text{for } \gamma = 0 \\ n \ll 1 & \text{for } \gamma \neq 0 \\ \kappa > n \end{cases} \quad (4.33)$$

For $\gamma = 0$: there is no propagating or wave-like solution, only a spatially exponentially decaying amplitude similar to the type known for total reflection in the medium with the lower index of refraction

For finite γ : some light can penetrate into the medium, but the light is damped over a distance shorter than the wavelength.

Fig. 4.5. Real (a) and imaginary (b) parts of the complex index of refraction for zero and finite damping

4.4 The Spectra of Reflection and Transmission

- Now discuss the spectra of reflection and transmission:
- Assume a single interface between vacuum(or air) and the medium, and normal incidence.

$$R(\omega) = \frac{I_r}{I_i} = \frac{[n(\omega) - 1]^2 + \kappa^2(\omega)}{[n(\omega) + 1]^2 + \kappa^2(\omega)}$$

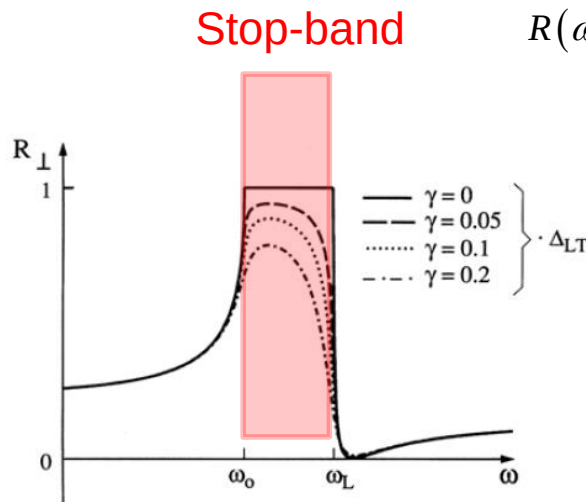


Fig. 4.6. The reflection spectrum of a single resonance with zero and finite damping for normal incidence

$$R(\omega) = \frac{(\epsilon_s^{1/2} - 1)^2}{(\epsilon_s^{1/2} + 1)^2} \quad \text{for } \omega_0 - \omega \gg \frac{1}{\hbar} \Delta_{LT}$$

R starts with an almost a constant value

$$R(\omega) = \frac{(\epsilon_b^{1/2} - 1)^2}{(\epsilon_b^{1/2} + 1)^2} \quad \text{for } \omega - \omega_0 \gg \frac{1}{\hbar} \Delta_{LT}$$

R converges to a lower constant value

- For vanishing damping, R increases just below ω_0 and reaches the value $R=1$ for $\omega=\omega_0$. Between ω_0 and ω_L $R=1$. For $\omega > \omega_L$ R drops rapidly and reaches 0 at the frequency where $n=1$. Then R increases again towards a constant value
- For finite damping the $R(\omega)$ is smoothed out due to the finite values of n for $\omega_0 \leq \omega \leq \omega_L$. The reflection minimum occurs also for weak damping only slightly above ω_L , but the reflection maximum is no longer related to ω_0

Reststrahlenbande

- The spectral region between ω_0 and ω_L is often called “Reststrahlenbande” for the following reason. If we send a light beam with a broad, essentially flat, spectral distribution of the intensity around ω_0 onto a sample and allow it to be reflected several times, e.g., in the configuration given in Fig. 4.7a, then significant intensity will remain as a “rest” only in the region

$$\omega_0 \leq \omega \leq \omega_L$$

- It should be noted that we do not assume any coherent superposition of the beams, i.e., the Reststrahlenbande comes only from $\varepsilon(\omega)$ and has nothing to do with Fabry–Perot resonators

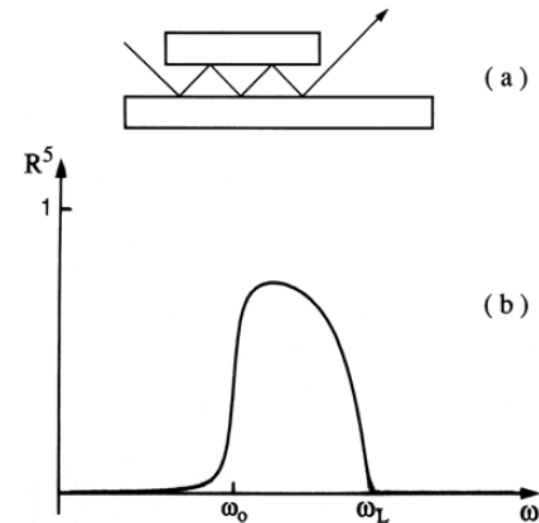


Fig. 4.7. An arrangement for multiple reflection from a medium (a) and the resulting “Reststrahlenbande” (b)

α_c

Semiconductor slab

- Handle semiconductor slabs with two surfaces and a geometrical thickness d
- (a) No interference
- (b) Interference

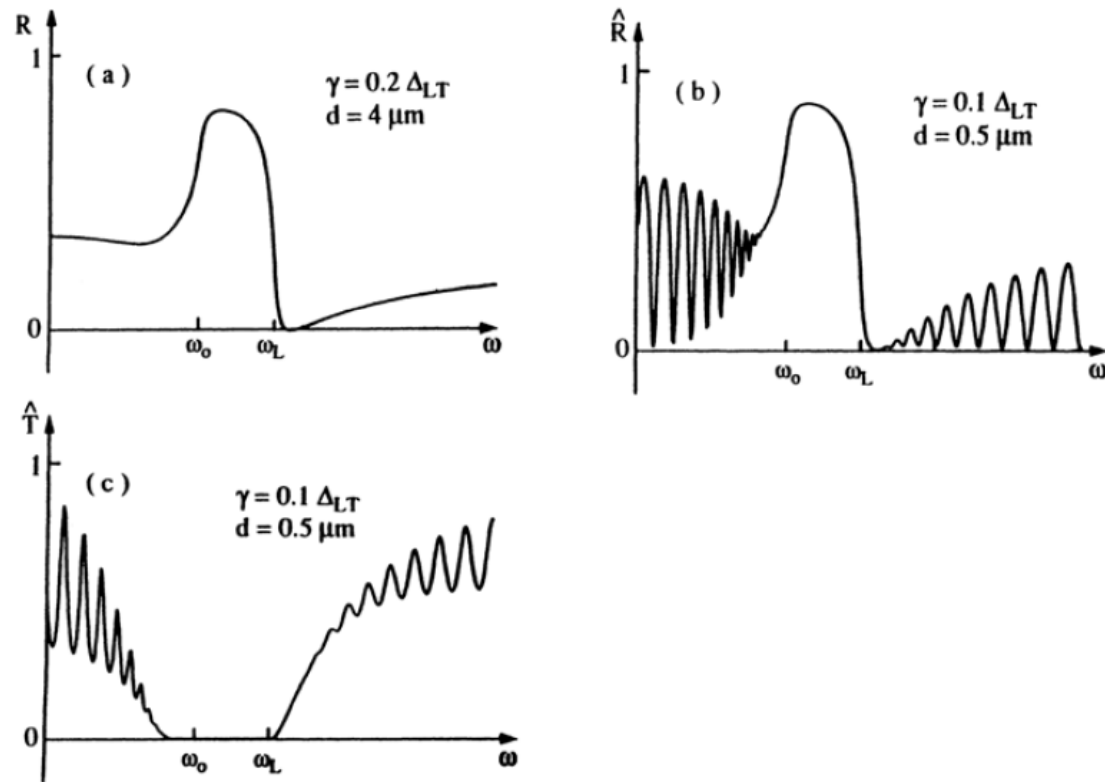


Fig. 4.8. The reflectivity $\hat{R}$ of a thin slab of matter in the vicinity of a resonance for the cases where intensities (a) or field amplitudes (b) have to be added and the transmission (b) for the case where amplitudes add

4.5 Interaction of Close Lying Resonances

- Discuss what happens for two close-lying resonances.
For instance :for the A and B Γ_5 excitons in ZnO.
- We consider two resonances with equal oscillator strength f and slightly $\omega_A < \omega_B$.
and we neglect damping and spatial dispersion.

$$\varepsilon_A(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_A^2 - \omega^2)}, \quad (4.37a)$$

$$\varepsilon_B(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_B^2 - \omega^2)}, \text{ and} \quad (4.38a)$$

$$\varepsilon_{\text{tot}}(\hbar\omega) = \varepsilon_b + \frac{f}{\hbar^2(\omega_A^2 - \omega^2)} + \frac{f}{\hbar^2(\omega_B^2 - \omega^2)} \quad (4.38b)$$

$$= \varepsilon_A(\hbar\omega) + \varepsilon_B(\hbar\omega) - \varepsilon_b \quad (4.38c)$$

The situation becomes significantly different for (4.38c) shown in Fig.4.9c.

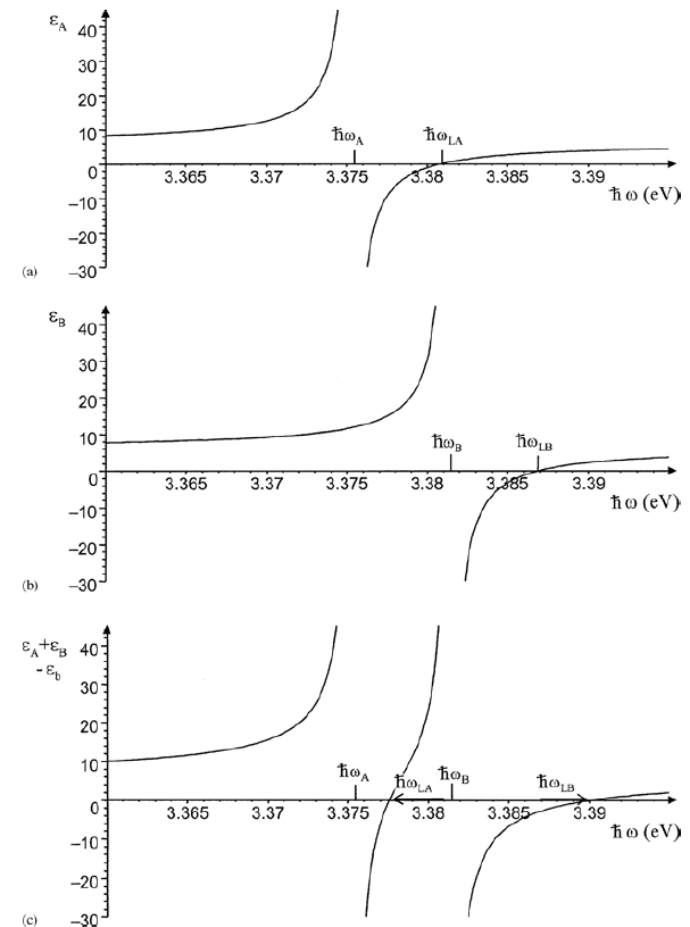


Fig. 4.9. The real part of $\varepsilon(\hbar\omega)$ for two resonances separately (a, b) and for two close lying resonances (c). Numbers on the x-axis refer roughly to ZnO